SPACE ROBOTICS LABORATORY · TOHOKU UNIVERSITY · PHASE 0

Moonwalk 2.0

Report 01 · Site, block and robot–block interface

Phase 0 · the base site at the lunar south pole, one osteomorphic block family sized to the robot, and three ways for the robot to grip it



TOWARDS A PERMANENT [illegible] SINCE ON THE MOON

The SPACE ROBOTICS LAB

SPACE ROBOTICS LABORATORY
TOHOKU UNIVERSITY · SENDAI

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RESEARCH STAY · PROF. KAZUYA YOSHIDA
SEPTEMBER 2026

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Robots build the outpost's whole shell from one family of regolith blocks, before the crew arrives. This first report sets the ground rules: what the south-pole site asks of the block, the block itself, the build phases, the literature behind the block, the robot–block interface and how it will be tested in simulation.

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OSTEOMORPHIC BLOCK FAMILY SIZED TO THE ROBOT

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WHAT TO ASK THE LAB · 11-WEEK PLAN

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_01

INTERACTIVE PARTS
3D moon tour · block explorer · grip animations · results dashboard

01



CHAPTER 01 · MOONWALK 2.0 · REPORT 01

The site

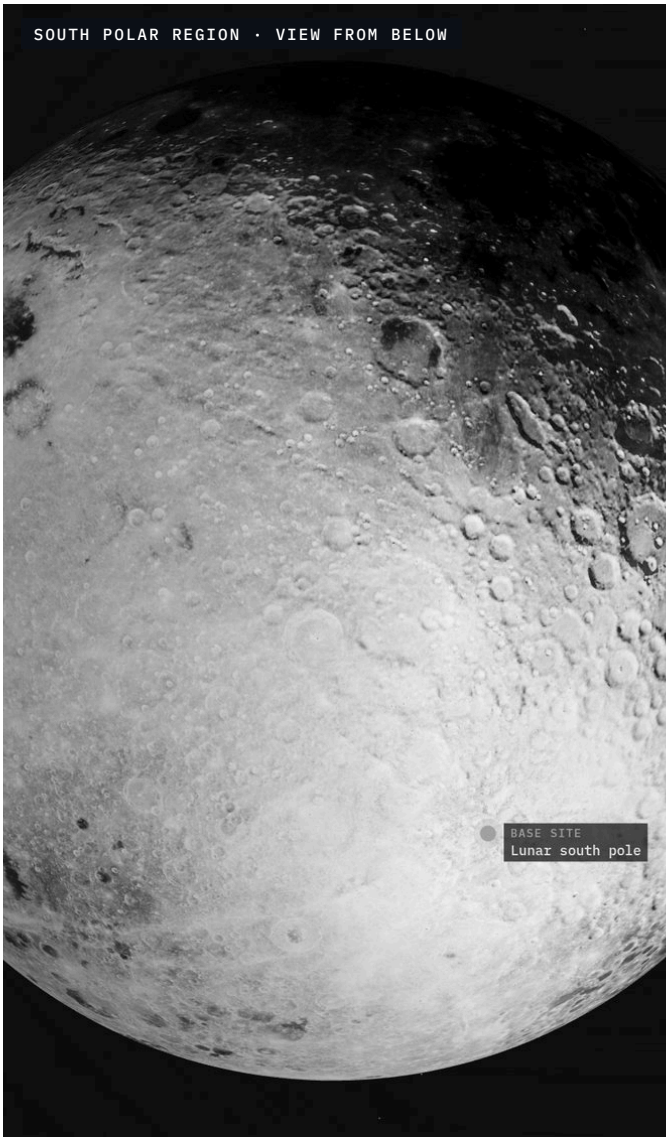
The south pole sets the rules

At the pole the Sun never rises more than about 1.5° above the horizon. Light arrives sideways and long shadows shape the terrain. Every structure in the programme answers to that low light.

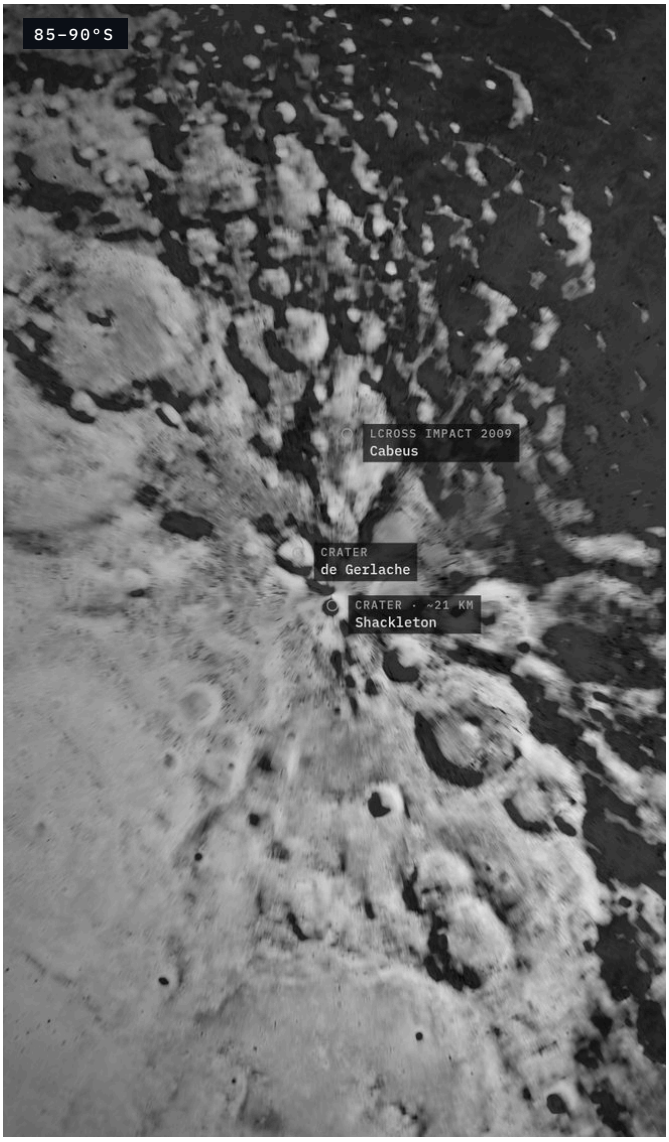
BASE SITE · LUNAR SOUTH POLE

Towards a permanent human presence on the Moon

Robots build the outpost's whole shell from one family of regolith blocks, before the crew arrives. The site decides what that block must be.



THE BASE SITE AT THE LUNAR SOUTH POLE



SHACKLETON, DE GERLACHE AND CABEUS (LCROSS IMPACT, 2009)

The lunar south pole

The Sun never rises more than about 1.5° above the horizon. Light arrives sideways, and long shadows shape the terrain.

Ridges that stay lit

Some crater rims and ridges, such as the ridge between Shackleton and de Gerlache, are sunlit for most of the year. That is where the power plant goes: vertical solar masts catching horizontal light.

Craters that never see the Sun

Crater floors in permanent shadow are among the coldest places measured in the solar system, and they hold water ice. NASA's LCROSS impactor confirmed water when it struck Cabeus in 2009.

WHAT THE SITE ASKS OF THE BLOCK · DESIGN INPUTS

1/6 g
Lift scales ~6×; block inertia does not

10–20°
Slopes the robots can handle

Dry-stacked
Extreme thermal swings crack bonded joints

Dust-tolerant
Fine, abrasive, charged regolith

01 · THE SITE

The south pole sets the rules

Every number below is an input to the block. The site is zoned so that every structure is a compressive block form and every distance is a block count.

LIGHT

Sun on the horizon

Solar collection on vertical masts, not ground panels. Quarry and print yard sit in the lit zone.

GRAVITY

1/6 g

Robot lift capacity scales about 6×. Compressive structures carry lower self-weight but the same tensile weakness.

SLOPES

10–20°

What the robots handle, by configuration. Haul paths stay below it; ramps are built from blocks.

THERMAL

Extreme swings

No bonded joints to crack. Guideway alignment checked across cycles.

REGOLITH

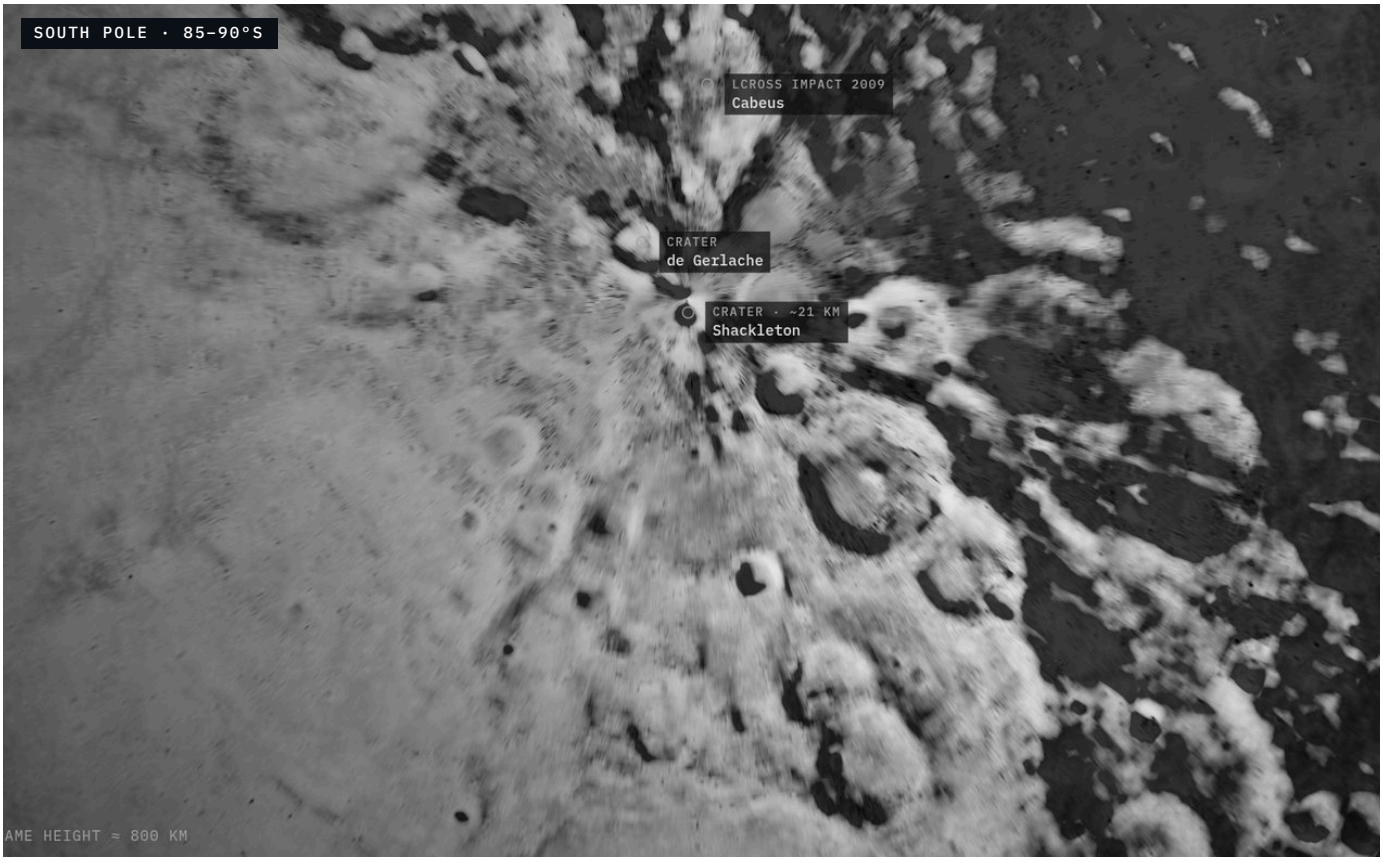
Abrasive, charged

Sealed robot joints, dust-tolerant grip surfaces, sintered pad face against plume erosion.

LINK

Earth–Moon delay

Supervised autonomy. The digital twin runs locally at the outpost.



3D MOON TOUR FROM THE LIVE REPORT: THE CAMERA TILTS TO THE POLE AND ZOOMS IN ON THE BASE SITE

SITE ZONING

Landing pad

Downrange of everything, on the flattest ground, with a berm ring toward the outpost.

Power plant

On the ridge with the longest illumination; storage in a shielded hollow beside it.

Corridor

The straightest line from pad to habitat. Guideway alignment is the tightest tolerance on site.

Habitat vault

Behind natural terrain at the corridor's far end, for free shielding.

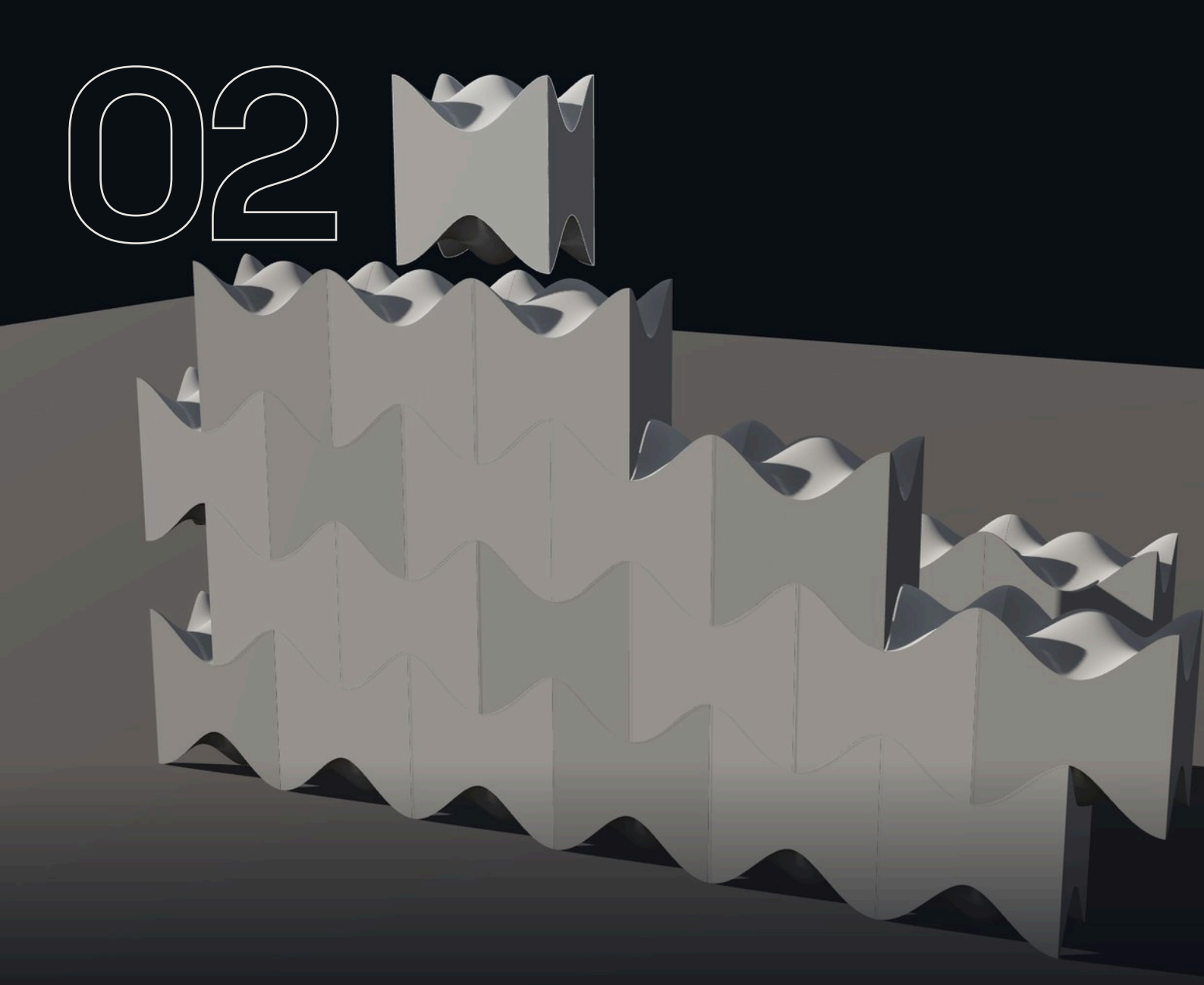
Logistics bays

Along the corridor, where the pod stops.

Quarry and print yard

Next to the power plant: the shortest haul from regolith to block to structure.

02



OSTEOMORPHIC BLOCK FAMILY IN RUNNING BOND · RENDER

CHAPTER 02 · MOONWALK 2.0 · REPORT 01

The block

One family, sized to the robot

A redesign of the lab's block as an osteomorphic block (Dyskin et al. 2003, 2005): two matching curved faces, the bottom one shifted by half a length, so each course self-aligns in running bond and holds without mortar.

02 · THE BLOCK

One family, sized to the robot

Two matching curved faces, the bottom one shifted by half a length, so each course self-aligns in running bond and holds without mortar under a peripheral constraint. Grades change density, finish and seat, never the interlock.

Compression only

Regolith has almost no tensile strength. Pavers, berms, piers, arches, vaults. No mortar, no adhesive.

Self-adjusting

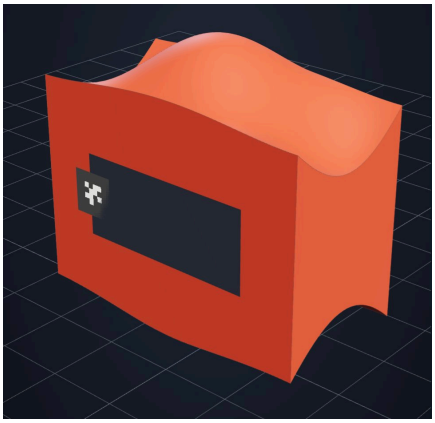
Concave meets convex, so a slightly misplaced block slides into place. Low-precision robots and in situ printing are enough.

Sized to the robot

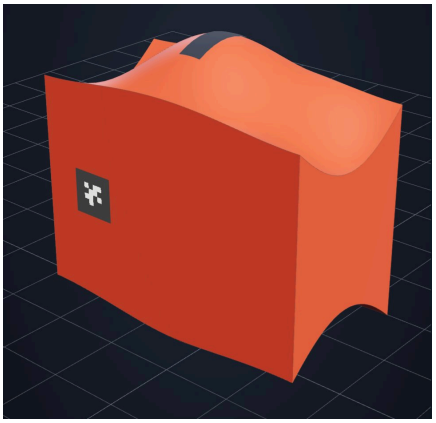
Mass is set by manipulator payload and inertia, not by the structure.

Needs a constraint

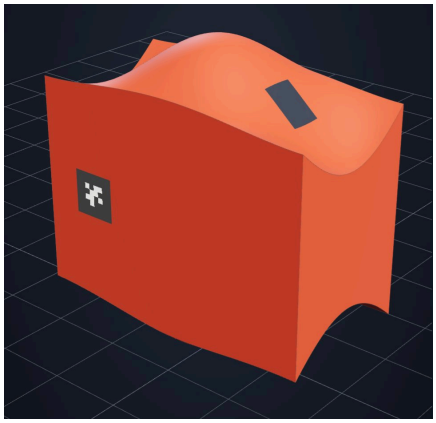
The assembly holds only when confined: by a frame, the terrain, tendons or its own weight.



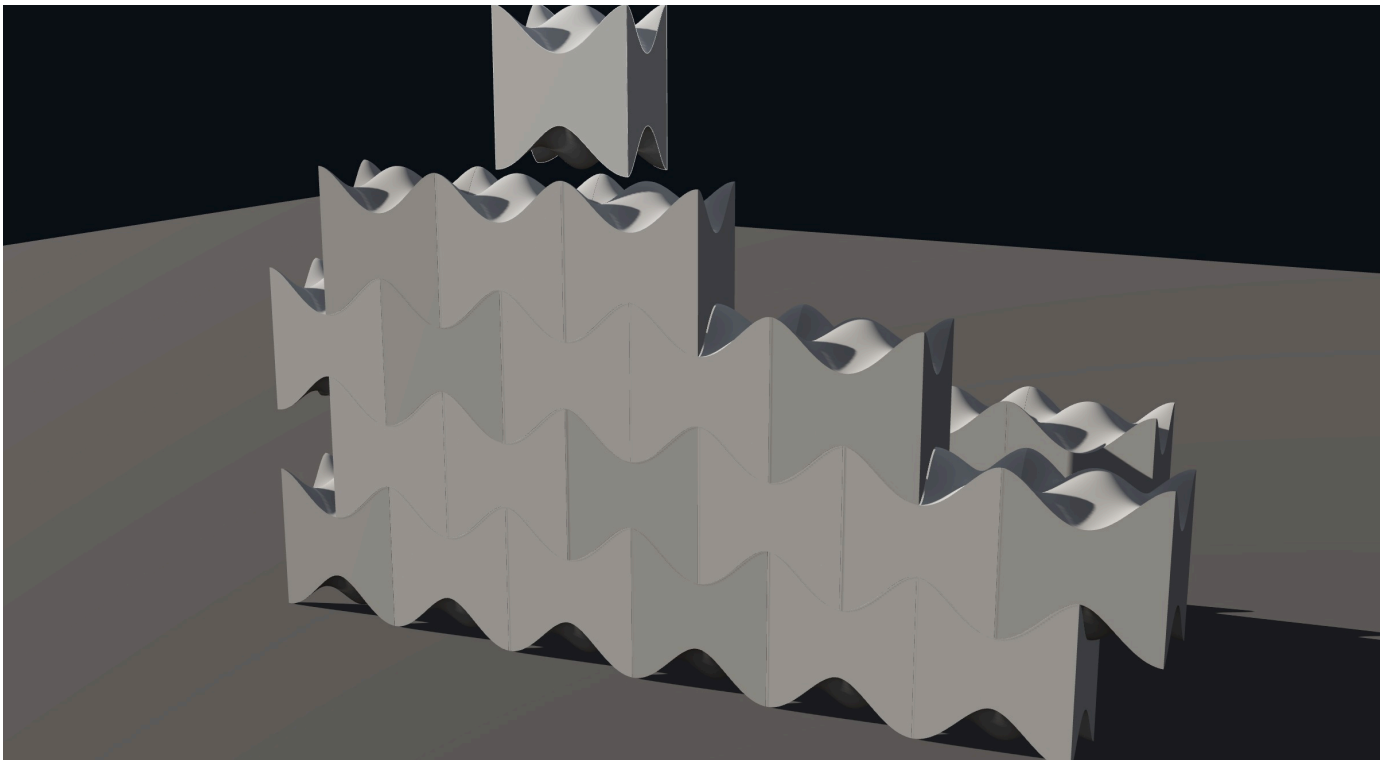
INTERFACE 1 · FLUSH SIDE HANDLE



INTERFACE 2 · TOP TWIST-LOCK SLOT



INTERFACE 3 · TOP POCKET GRIP



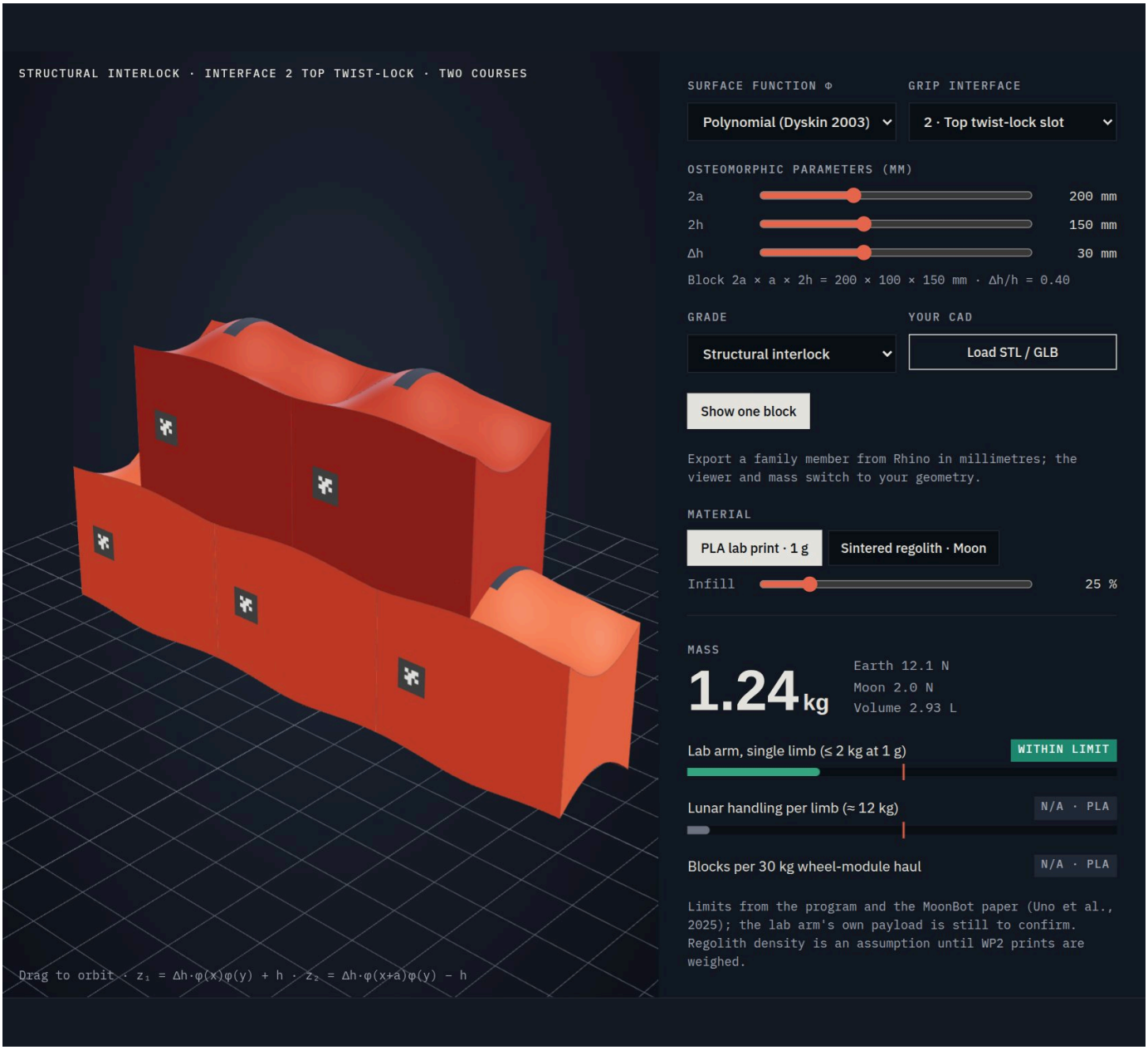
RUNNING BOND: EACH COURSE NESTS IN THE CURVED TOP OF THE COURSE BELOW; NO MORTAR

GEOMETRY · DYSKIN ET AL. 2003

$z_1 = \Delta h \cdot \varphi(x)\varphi(y) + h$ · $z_2 = \Delta h \cdot \varphi(x+a)\varphi(y) - h$. The surface function φ is polynomial (Dyskin 2003) or cosine (Yong 2011). Each block carries a fiducial marker on its side face for pose checks.

BLOCK EXPLORER · FROM THE LIVE REPORT

Parameters, mass and payload checks



INTERACTIVE EXPLORER: SURFACE FUNCTION, GRIP INTERFACE, OSTEOMORPHIC PARAMETERS, GRADE, MATERIAL AND PAYLOAD GAUGES

Default member

2a × a × 2h = 200 × 100 × 150 mm, Δh = 30 mm (Δh/h = 0.40), structural interlock grade, two courses in running bond.

Lab print

PLA at 25 % infill: **1.24 kg** (12.1 N on Earth, 2.0 N on the Moon, 2.93 L). Within the lab arm's single-limb limit of 2 kg at 1 g.

On the Moon

Sintered regolith is checked against lunar handling per limb (≈ 12 kg) and the number of blocks per 30 kg wheel-module haul.

Grades

- **Structural interlock** · walls, piers, arches
- **Paver** · landing pad surface
- **Shielding bulk** · berms and cover
- **Rail-seat** · guideway supports

Importing CAD

Any family member exported from Rhino in millimetres (STL or GLB) can be loaded into the explorer, which then shows that geometry and recalculates its mass.

Limits from the programme and the MoonBot paper (Uno et al., 2025); the lab arm's own payload is still to confirm. Regolith density is an assumption until WP2 prints are weighed.

03



SHACKLETON–DE GERLACHE RIDGE · LROC · NASA/GSFC/ARIZONA STATE UNIVERSITY

0	1	2	3	4
GROUND WORK	**LANDING PAD**	**POWER PLANT**	**SHELL**	**PLUG-INS**
Block, prints, robot demo, twin	Sintered pavers, berm ring	Solar masts on block footings	Guideway, vault, bays	Pods docked into the shell

CHAPTER 03 · MOONWALK 2.0 · REPORT 01

Phases

The shell first, then the plug-ins

Phases 1 to 3 are built entirely from blocks by robots, powered by the outpost's own plant. Phase 4 docks systems made on Earth into the finished shell. Each phase hands a capability to the next.

03 · PHASES

The shell first, then the plug-ins

Phases 1 to 3 are built entirely from blocks by robots, powered by the outpost's own plant. Phase 4 docks systems made on Earth into the finished shell. Each phase hands a capability to the next.

0

WE ARE HERE

Ground work

Block design, simulant prints, robot stacking demo, digital twin of the full programme.

Robot stacks a pier or arch unattended.

1

Landing pad

Sintered pavers plus an ejecta berm ring. Every later landing needs it.

Survives repeated plume loading.

2

Power plant

Solar masts on block footings, shielded storage, power trunk in block-lined trenches.

Robots run on outpost power.

3

Shell

Maglev piers and guideway, open habitat vault, logistics bays, all with standard interfaces.

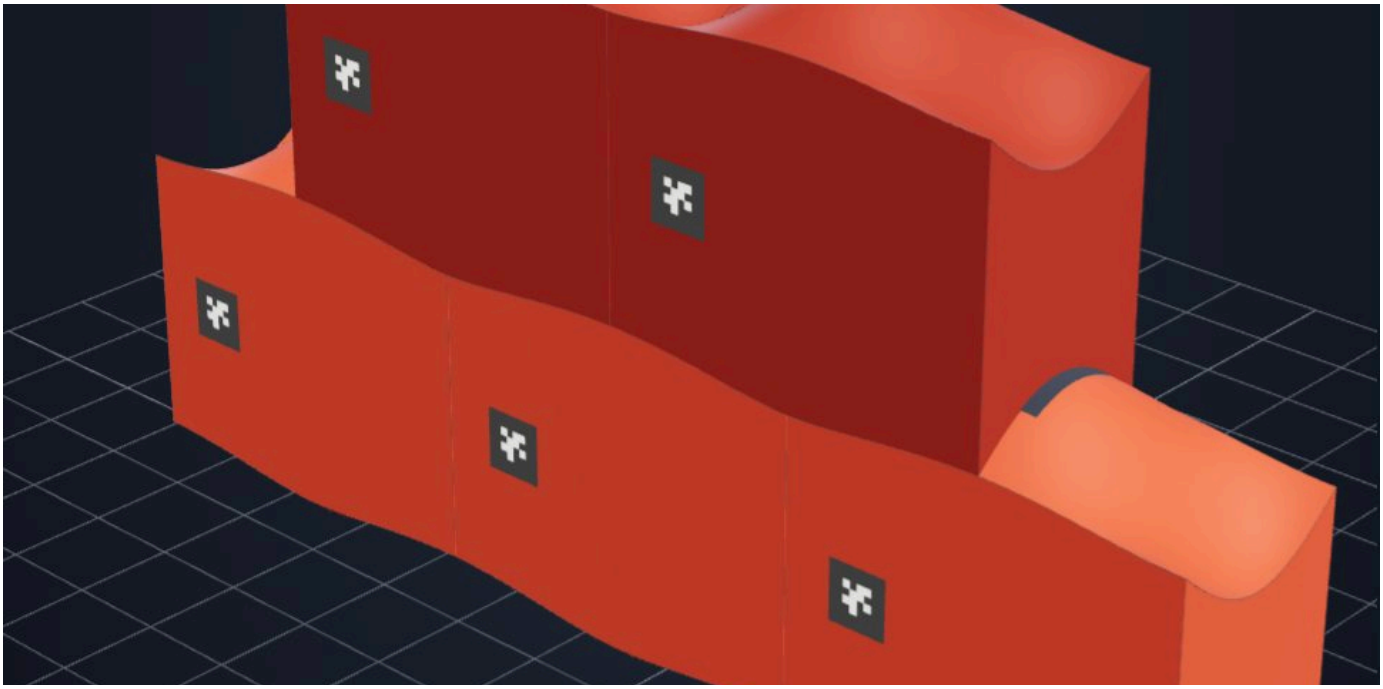
Verified against the twin.

4

Plug-ins

Robot pod, then cargo, then crew pod and inflatable habitat. The vault closes after insertion.

Docks without shell rework.



PHASE 0 · THE BLOCK FAMILY IN TWO COURSES, RUNNING BOND: THE UNIT EVERY LATER PHASE IS BUILT FROM

PHASE	CAPABILITY HANDED ON	EXIT TEST
0 · Ground work	Block family, prints, stacking demo, digital twin	Robot stacks a pier or arch unattended
1 · Landing pad	Safe repeated landings	Survives repeated plume loading
2 · Power plant	Outpost power for the robots	Robots run on outpost power
3 · Shell	Guideway, vault and bays with standard interfaces	Verified against the twin
4 · Plug-ins	Pods and inflatable habitat docked into the shell	Docks without shell rework

04

ARCHITECTURE BOOKSHELF · PHOTO SUPPLIED BY THE AUTHOR



CHAPTER 04 · MOONWALK 2.0 · REPORT 01

Literature review

Topological interlocking

The papers the block design is built on, grouped by the question each one answers, with what each contributes to Moonwalk 2.0.

04 · LITERATURE REVIEW

Topological interlocking

The papers the block design is built on, grouped by the question each one answers. Each entry says what it contributes to Moonwalk 2.0.

OSTEOMORPHIC BLOCKS · THE GEOMETRY USED

ADV. ENG. MATERIALS 5(3) · 2003
PDF in project library

Fracture resistant structures based on topological interlocking with non-planar contacts

A. V. Dyskin, Y. Estrin, A. J. Kanel-Belov, E. Pasternak

Defines the osteomorphic block: $z_1 = \Delta h \cdot \varphi(x)\varphi(y) + h$, $z_2 = \Delta h \cdot \varphi(x+a)\varphi(y) - h$. The equations behind the block explorer.

PHD THESIS · UWA · 2011
PDF in project library

Utilisation of topologically-interlocking osteomorphic blocks for multi-purpose civil construction

D. Yong Hsien Ta (supervisor A. V. Dyskin)

Sine/cosine and circular-arc variants of the block, and one block for walls, paving and retaining walls. Source of the cosine φ option.

NEXUS NETWORK JOURNAL 26 · 2024
doi.org/10.1007/s00004-023-00756-7

Integration of design and performance evaluation for reusable osteomorphic-block masonry

K. Harsono, S.-G. Shih, F. Wagiri, W. Alfred

A design loop of contact interface, block generation, FEA and block set. The same sequence as WP1 to WP2.

TOPOLOGICAL INTERLOCKING AND CONSTRAINT

MATER. SCI. ENG. C 31 · 2011
doi.org/10.1016/j.msec.2010.11.011

Topological interlocking as a material design concept

Y. Estrin, A. V. Dyskin, E. Pasternak

The principle: elements held by shape and a peripheral constraint, with tolerance to missing blocks and crack arrest.

FRONT. STRUCT. CIV. ENG. 6(2) · 2012
doi.org/10.1007/s11709-012-0156-8

Mortarless structures based on topological interlocking

A. V. Dyskin, E. Pasternak, Y. Estrin

Constraint by frame, topography, tendons or self-weight. Tells us what holds a lunar course together.

J. MATER. RES. TECHNOL. 15 · 2021
doi.org/10.1016/j.jmrt.2021.08.064

Design of architectured materials based on topological and geometrical interlocking

Y. Estrin, V. R. Krishnamurthy, E. Akleman

Review from platonic solids to Delaunay lofts. The wider design space for the block family.

LECTURE · VIENNA · 2010
PDF in project library

Topological interlocking as a materials design concept

Y. Estrin

Slides on interlocked cubes, octahedra and osteomorphic blocks, with their mechanical response.

EXTRATERRESTRIAL CONSTRUCTION, FABRICATION AND TESTING

ACTA ASTRONAUTICA 57 · 2005
doi.org/10.1016/j.actaastro.2004.12.005

The principle of topological interlocking in extraterrestrial construction

A. V. Dyskin, Y. Estrin, E. Pasternak, H. C. Khor, A. J. Kanel-Belov

Osteomorphic bricks sintered from regolith. They self-adjust, so low-precision robots can place them. The case for Moonwalk's block.

MATERIALS & DESIGN 254 · 2025
sciencedirect.com/science/article/pii/S0264127525004691

3D concrete printing of topological interlocking blocks

T. Goertzen, T. Neef, P. Scheffler, D. Macek, V. Mechtcherine, A. C. Niemeyer

Printed interlocking blocks in a fixed frame. A precedent for the print route and the frame as constraint.

MATERIALS AND DESIGN 186 · 2020
sciencedirect.com/science/article/pii/S0264127519307853

Mechanical behaviour of composite structures made of topologically interlocking concrete bricks with soft interfaces

A. Rezaee Javan, H. Seifi, X. Lin, Y. M. Xie

Effect of confining load and interface friction. Feeds the dust and friction cases in simulation.

IEEE TRANS. FIELD ROBOTICS · 2025
doi.org/10.1109/TFR.2025.3624346

MoonBot

K. Uno et al., Space Robotics Lab, Tohoku University

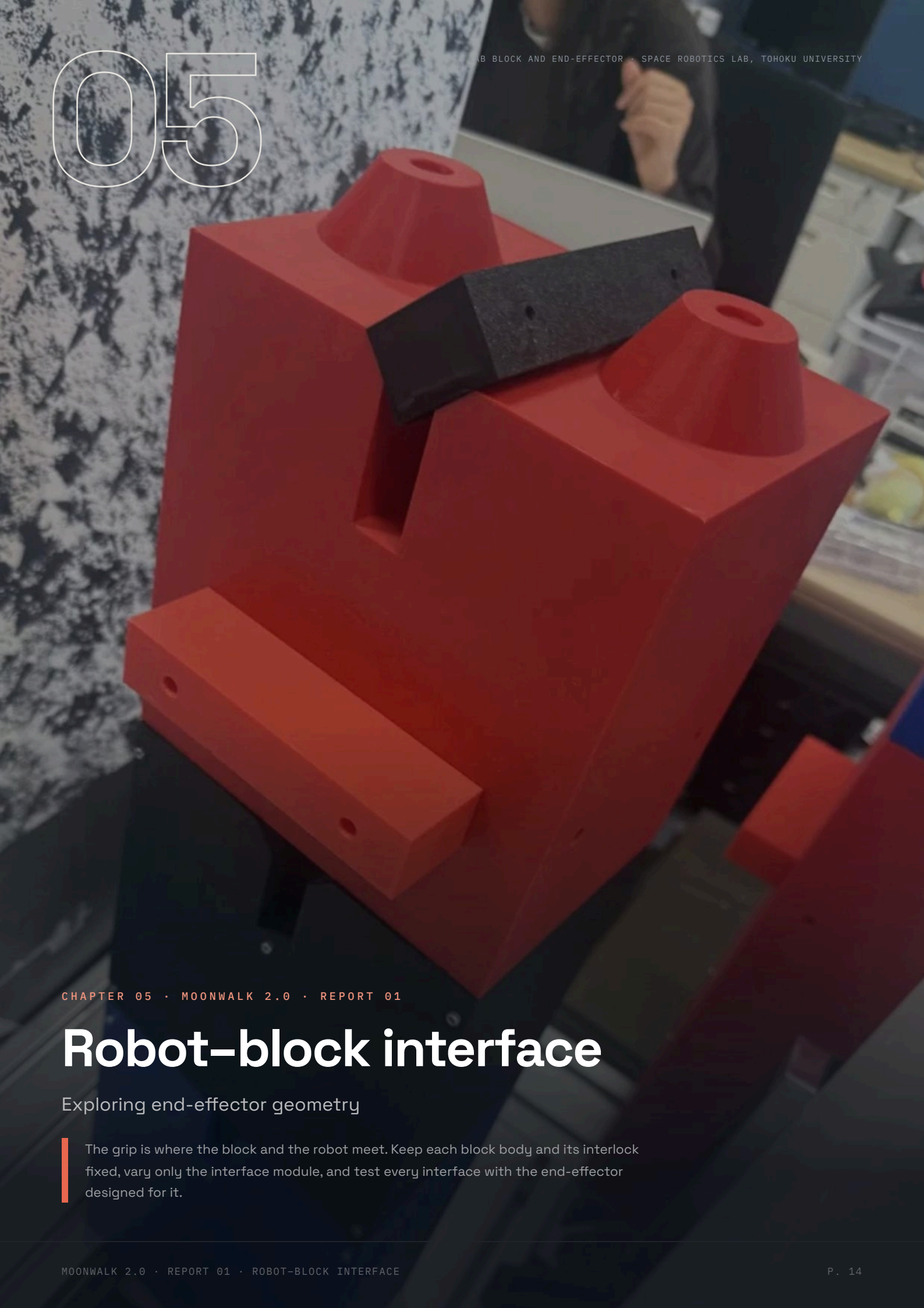
The reference robot from the programme: payload, gripper and configurations set block mass and grip limits.

HOW THE CLUSTERS FEED THE WORK

- **Geometry** → the block explorer and the Grasshopper block family (WP1)
- **Constraint** → what confines each structure: frame, terrain, tendons or self-weight (WP3)
- **Fabrication** → print route and frame as constraint (WP2)
- **Testing** → friction and dust cases, robot payload limits (WP4)



WHERE THE BLOCKS GO: THE SOUTH-POLE TERRAIN AROUND SHACKLETON AND DE GERLACHE



CHAPTER 05 · MOONWALK 2.0 · REPORT 01

Robot–block interface

Exploring end-effector geometry

The grip is where the block and the robot meet. Keep each block body and its interlock fixed, vary only the interface module, and test every interface with the end-effector designed for it.

05 · ROBOT–BLOCK INTERFACE

Exploring end-effector geometry

The method: keep each block body and its interlock fixed, vary only the interface module, and test every interface with the end-effector designed for it.

Step 1 · now
Lab block today
Two cone locators on top, a bar handle standing off one side, gripped by the lab's C-jaw.

Step 2 · now
Three interface options
One compatible with the lab jaw, two gripping from above.

Step 3 · WP4
Test in Isaac Sim
All three on the same blocks, same tasks, same metrics.

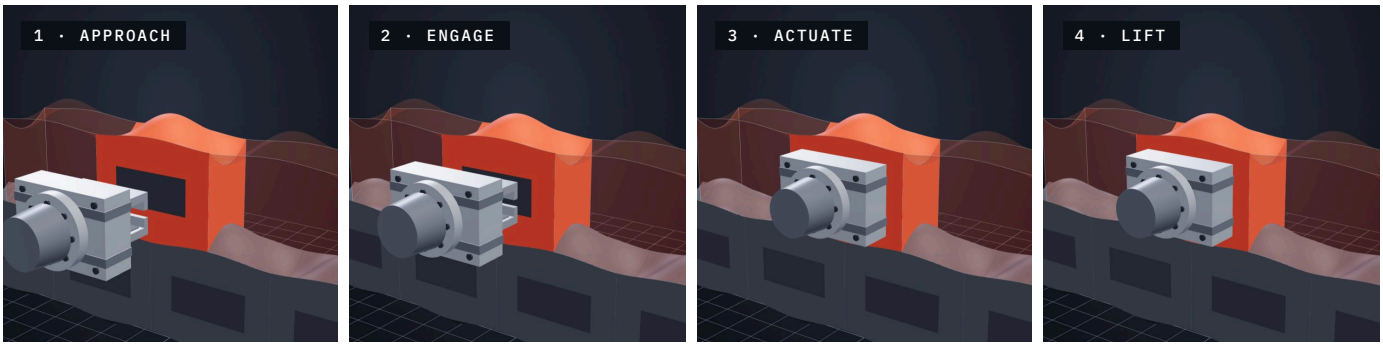
Step 4 · after
Select, print, lab test
The best interface is printed and tried on the real arm, as the lab allows.

OPTION 1 · LAB-COMPATIBLE

Flush side handle + C-jaw

Existing lab end-effector (C-jaw on a bar). The handle is built into the printed body and recessed flush, so it matches the existing jaw without sticking out.

Approach **horizontal**
Actuation **jaw closes**
Side access **front face free**
Hardware **existing**

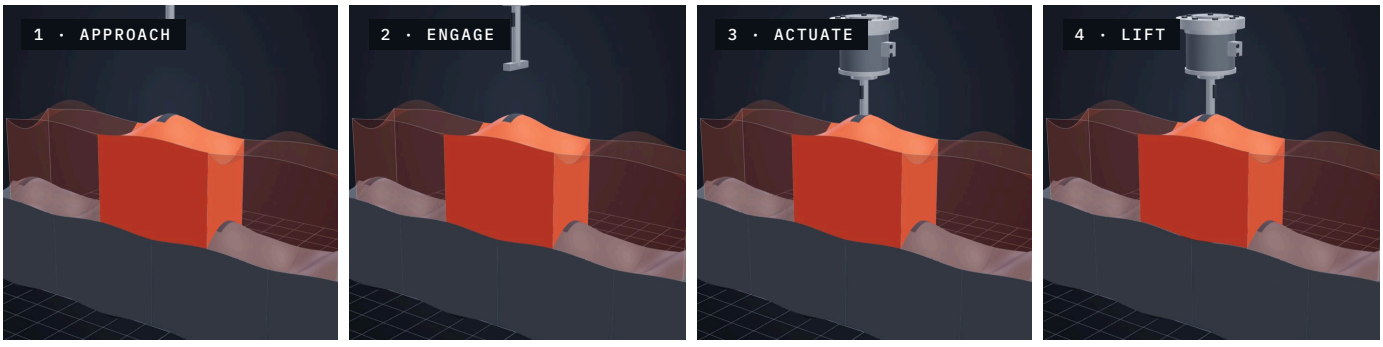


OPTION 2 · TOP TWIST-LOCK

Key into slot, turn 90°

New end-effector: a T-key enters the top slot and turns 90° under the undercut. The slot and cavity carry straight over from the lab model.

Approach **vertical**
Actuation **wrist turn 90°**
Side access **none**
Hold **form lock**



OPTION 3 · TOP POCKET GRIP

Two fingers spread outward

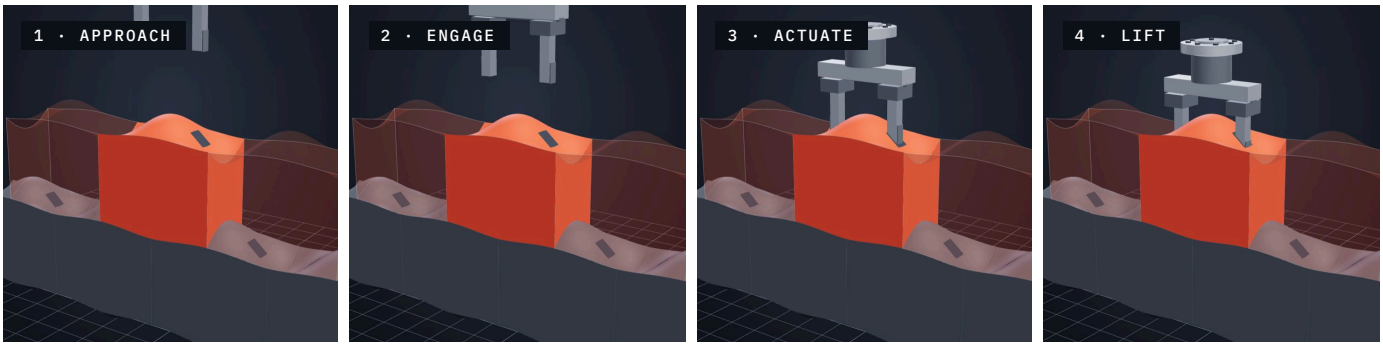
New end-effector: two fingers on a linear slide enter pockets in the free top corners and spread against the pocket walls.

Approach **vertical**

Actuation **fingers spread**

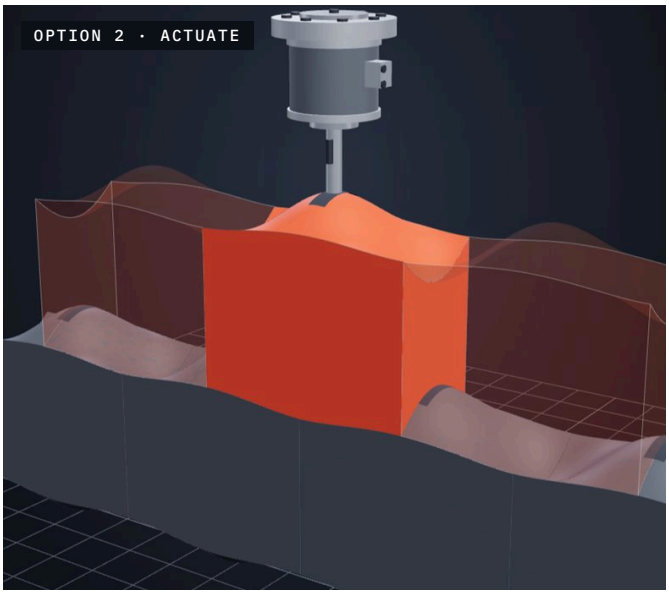
Side access **none**

Hold **friction + wall**

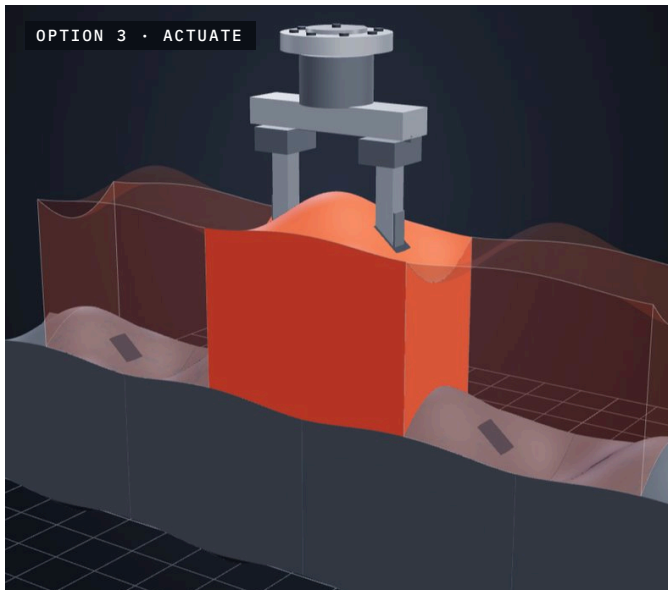


Why gripping from above matters

Options 2 and 3 grip from above. One end-effector then works for walls, columns and plates, because the top face is free in all of them, and blocks can be placed tightly side by side with no room needed for side access. Option 1 stays as the baseline: it runs on the lab's current hardware.



T-KEY TURNED 90° UNDER THE UNDERCUT



FINGERS SPREAD AGAINST THE POCKET WALLS

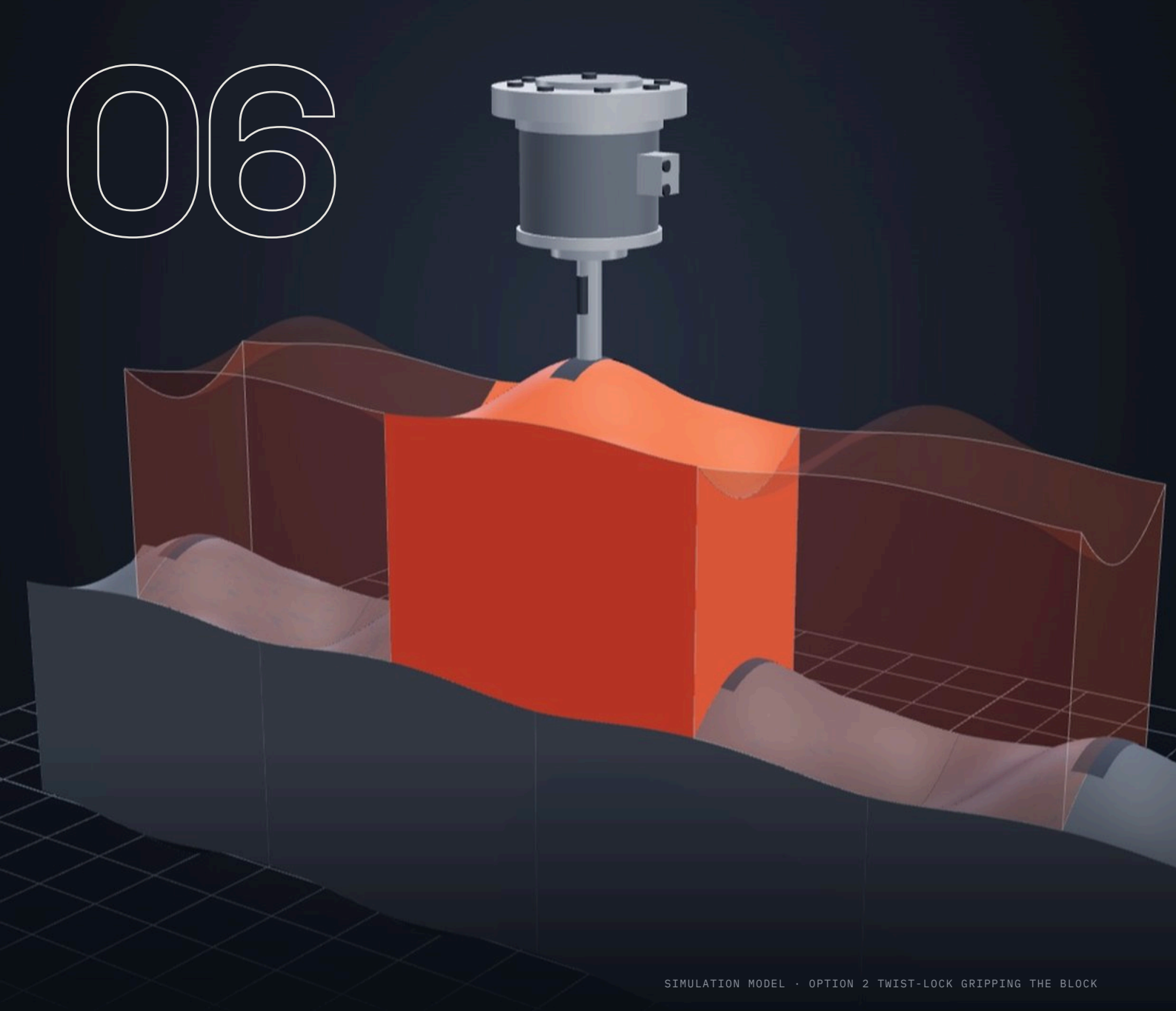
HOW THE OPTIONS ARE ORGANISED

One Grasshopper definition, one switch

The block body and its interlocking features stay the same for all options. The grip is a separate interface module, chosen with a Value List (a drop-down). Every block in the family can then carry any of the three interfaces.

INPUTS	FIXED PER MEMBER	VALUE LIST	FAMILY	OUTPUTS
Shared parameters	**Body + interlock**	**Interface module**	**Every member**	**STL + USD**
Envelope, mass, interlock depth, print tolerance	Plate, column, wall or voussoir	1 · side handle · 2 · twist-lock · 3 · pocket grip	Carries the chosen interface	Print files for the desktop printer, assets for Isaac Sim

06



SIMULATION MODEL · OPTION 2 TWIST-LOCK GRIPPING THE BLOCK

CHAPTER 06 · MOONWALK 2.0 · REPORT 01

Simulation

Testing all three in Isaac Sim

Each end-effector is modelled as a simple gripper and tested on the same blocks, through the same tasks, with the same metrics. The dashboard reads the simulation logs directly.

06 · SIMULATION

Testing all three in Isaac Sim

Each end-effector is modelled as a simple gripper and tested on the same blocks, through the same tasks, with the same metrics.

Grip reliability

Pick, lift and carry: does the block stay held through acceleration and turns?

Placement precision

Position error after seating on the course below, in mm.

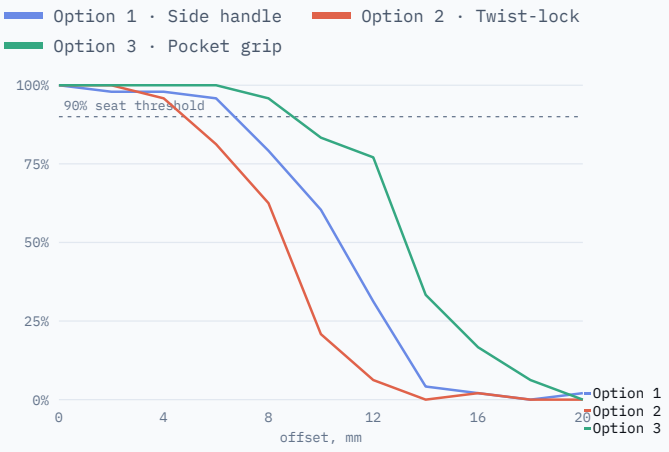
Misalignment

Start offset stepped up until the block no longer seats. The limit is the capture range.

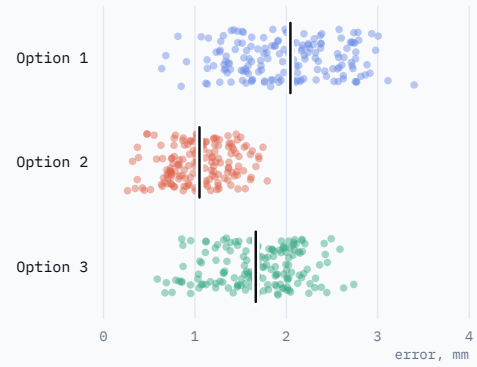
Dust

Lower friction and partly filled features, to see which interface degrades first.

ILLUSTRATIVE DATA · NOT RESULTS · the dashboard layout is shown with placeholder data until the Isaac Sim CSV is loaded



GRIP SUCCESS VS START OFFSET · % OF TRIALS SEATED



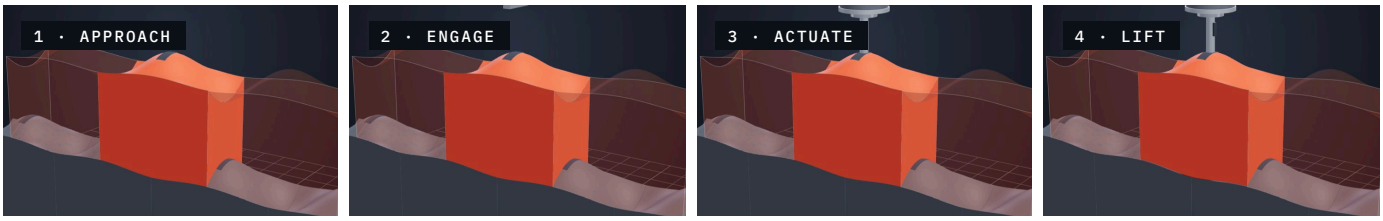
PLACEMENT ERROR · SUCCESSFUL TRIALS · BAR = MEDIAN

RESULTS CSV FORMAT

`option (1–3), member (plate | column | wall | arch), condition (nominal | dust | lunar), offset_mm, success (0/1), place_err_mm`

Filters: condition (nominal, dust, lunar g) and family member (plate, column, wall, arch voussoir).

ONE TRIAL · THE SAME FOUR STEPS FOR EVERY OPTION



RESEARCH OUTCOME

A measured comparison of three robot–block interfaces

How much placement tolerance each interface buys, and at what cost in hardware. The result feeds the robot–block section of the paper and tells the lab which interface to print and try on the real arm.

WP4 · weeks 7–9 of the 11-week plan · learning Isaac Sim in parallel now

07



Q1

Can it change?

Can the end-effector be modified or replaced?

Q2

Jaw dimensions

Dimensions and clamping range of the current jaw

Q3

Payload

The arm's payload limit, which caps block mass

Q4

CAD file

A CAD model of the end-effector for simulation

11-WEEK PLAN · MOONWALK 2.0 PROGRAMME · WP1–WP5

CHAPTER 07 · MOONWALK 2.0 · REPORT 01

Next steps

What to ask the lab

> Four answers decide whether option 1 stays the baseline only, or whether options 2 and 3 can go onto the real arm.

07 · NEXT STEPS

What to ask the lab

Four answers decide whether option 1 stays the baseline only, or whether options 2 and 3 can go onto the real arm.

Q1
Can it change?
Can the end-effector be modified or replaced?

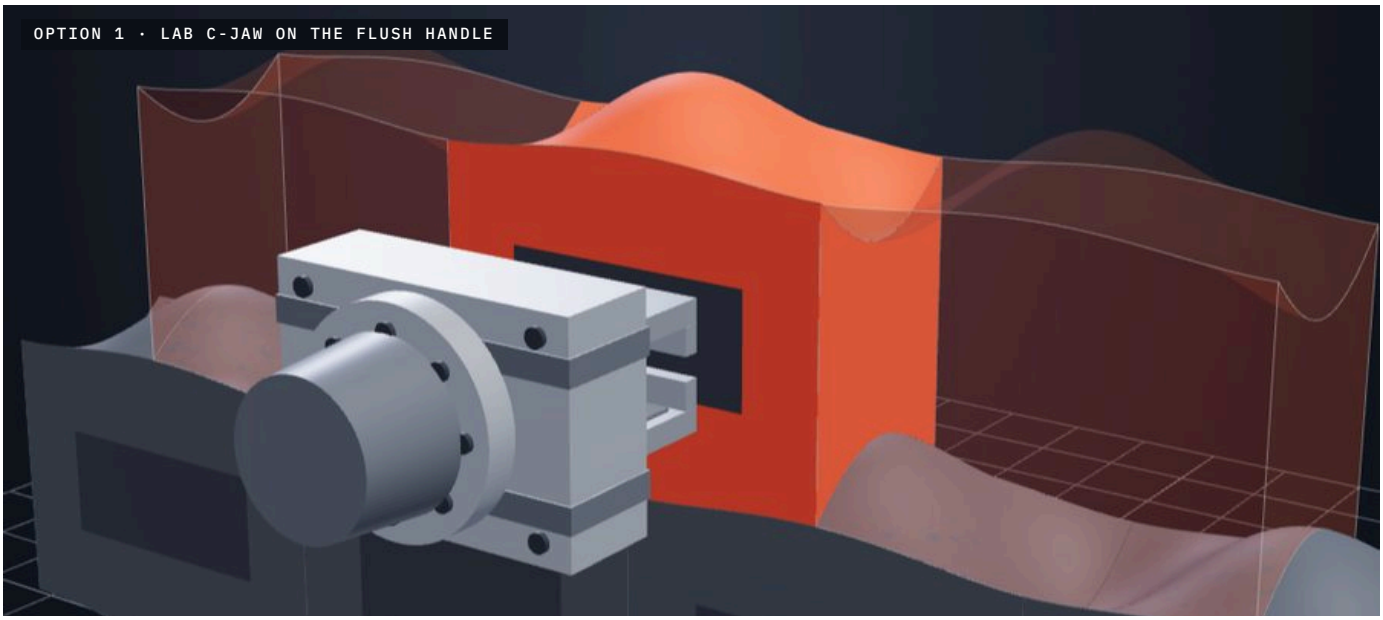
Q2
Jaw dimensions
The current jaw's dimensions and clamping range.

Q3
Payload
The arm's payload limit, which caps block mass in the explorer.

Q4
CAD file
Whether a CAD file of the end-effector exists, for the simulation model.

11-WEEK PLAN

WEEKS	FOCUS	MILESTONE
1–2	WP1: constraints fixed, parametric block family modelled, first PLA prints	Block family frozen in CAD
3–4	WP2: desktop-printed geometry checks, fiducial legibility, iterate and freeze	Geometry checked on printed parts
5–6	WP3: structures and plug-in interfaces designed from the block family	Interface specification issued
7–9	WP4: Isaac Sim set up, robot models imported, three interfaces tested	Robot places blocks in simulation; limits documented
10–11	WP5: full build simulation, animation, paper draft	Animation and paper draft delivered



Q1–Q4 DECIDE WHETHER THIS LAB-COMPATIBLE OPTION STAYS THE ONLY ONE ON THE REAL ARM

NOW
WP1 · block family modelled in Grasshopper, first PLA prints of the three interfaces.

NEEDS FROM THE LAB
Answers to Q1–Q4, so the explorer limits and the simulation gripper match the real arm.

NEXT REPORT
Structures from the block family and the first robot stacking tests.



The SPACE ROBOTICS LAB

Moonwalk 2.0

REPORT 01 · SITE, BLOCK AND ROBOT–BLOCK INTERFACE

LIVE REPORT
arhimatrix.github.io/arhimatrix-moonwalk-2-rep_01

INTERACTIVE PARTS
3D moon tour · block explorer · grip animations · results dashboard

REPORTS
Report 01 · 02 · 03 · 04 at arhimatrix.github.io

SOURCES
Moon map: Celestia (Clementine/NASA) · relief: Stellarium · 3D: three.js

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